HEALTHCONNECT EXPERIENCE LAB

Project Title: Healthconnect Clinic Project Summary Report
Program: AnalystLab Africa Data Analytics Internship
Week 5

Business Question: **"**How can HealthConnect Clinic use data to reduce missed appointments and improve the patient support experience?"

Transition / Foundation: Week 5 built directly on the data preparation, business-question definition and KPI planning completed during Week 4. The Week 4 foundation established the primary business objective of understanding appointment attendance and identifying factors contributing to missed appointments. Based on this foundation, Week 5 focused on transforming the prepared HealthConnect appointment dataset into actionable insights through Exploratory Data Analysis (EDA), KPI development and interactive Power BI dashboards.

The analysis prioritised appointment characteristics, patient characteristics, previous appointment history, reminder practices, waiting time, distance to the clinic, cancellations and appointment outcomes.

Data Preparation: The HealthConnect appointment dataset was reviewed and prepared for analysis in Power BI.

Key preparation activities included:

- Reviewed appointment, patient, reminder and outcome fields for consistency.
- Confirmed categorical variables such as: Appointment type, Appointment time, Appointment Day, Appointment outcome, Reminder channel & Age group.
- Created Distance Group categories to support analysis of distance and no-show behaviour.
- Used waiting-time groupings to examine appointment volume across waiting-time ranges.
- Developed calculated measures for key performance indicators.
- Reviewed missing values in numerical fields.
- Identified 60 blank values in Waiting Time and 90 blank values in Distance to Clinic.
- Identified negative values in Booking Lead Time due to inconsistencies in the demonstration dataset.

Decision regarding Booking Lead Time: Booking Lead Time was excluded from the Week 5 analysis because the dataset contained negative values that could produce misleading conclusions. This decision reduced the risk of presenting inaccurate findings while keeping the analysis focused.

Exploratory Data Analysis: EDA was conducted in Power BI to investigate relationships between appointment outcomes and key patient/support factors.

The main areas analysed were:

1. Appointment characteristics;
   Appointment Type
   Appointment Time
   Appointment Day
   Appointment Outcome

2. Patient characteristics;
   Age Group
   Distance to Clinic

3. Previous appointment behaviour;
   Previous No-Shows
   Previous No-Shows by Appointment Type

4. Patient support;
   Reminder Channel
   Reminder Coverage
   Appointment Outcome by Reminder Channel

5. Appointment operations;
   Waiting Time
   Waiting Time by Appointment Outcome
   Waiting Time distribution
   Cancellation behaviour

The analysis was designed to identify patterns that could help HealthConnect Clinic understand who is more likely to miss appointments, when missed appointments occur, and where patient-support interventions may be most useful.

KPI Development: Four primary KPIs were selected for the executive dashboard:

S/N	KPI	Result	Purpose
1.	No-Show Rate	48.5%	Measures the proportion of appointments missed by patients.
2.	Cancellation Rate	5.3%	Measures the proportion of appointment cancelled.
3.	Reminder Coverage	72.7%	Measures the proportion of appointments receiving a reminder.
4.	Average Waiting Time	24.19mins	Provides an indicator of the patient appointment experience.

These KPIs provide a balanced view of attendance, cancellations, patient support and operational experience.

Dashboard Visualizations: A three-page interactive Power BI dashboard was developed.

Executive Appointment Overview (Page 1): The executive page is designed to allow decision-makers to quickly understand overall appointment performance and identify areas requiring attention.

- No-Show Rate KPI
- Cancellation Rate KPI
- Reminder Coverage KPI
- Average Waiting Time KPI
- No-Show Rate by Reminder Channel
- No-Show Rate by Appointment Day
- Monthly Appointment Volume by Outcome
- No-Show Rate by Appointment Type
- Appointment Outcome Distribution

No-Show Drivers (Page 2): This page focuses specifically on factors associated with missed appointments. The key visuals include:

- No-Show Rate by Age Group
- No-Show Rate by Appointment Type
- No-Show Rate by Distance to Clinic
- No-Show Rate by Appointment Time

- No-Show Rate by Appointment Type and Previous No-Shows

A matrix was particularly useful for examining the relationship between previous no-show behaviour and subsequent no-show rates across appointment types.

Patient Support & Appointment Operations (Page 3): This page provides an operational perspective on how reminder practices, waiting times and appointment characteristics relate to patient outcomes. The key visuals include:

- Appointment Outcome by Reminder Channel
- Appointment Outcome by Time of Day
- Cancellation Rate by Appointment Type
- Average Waiting Time by Appointment Outcome
- Appointment Volume by Waiting Time
- Reminder Coverage by Appointment Type

Key Findings: The EDA identified several important patterns.

1. Missed appointments are a major operational issue: The overall no-show rate is 48.5%, meaning almost half of recorded appointments resulted in a no-show. This indicates that appointment attendance is a significant area for intervention.
2. Reminder channel is associated with different no-show rates: The observed no-show rates were:
   None - 51.4%
   WhatsApp - 49.8%
   Email - 48.4%
   SMS - 45.8%

   This suggests that reminder coverage and channel strategy deserve further attention.
3. Previous no-show behaviour is an important indicator: No-show rates increased substantially as the number of previous no-shows increased. The analysis therefore suggests that previous attendance behaviour may be useful for identifying patients who require additional appointment support.
4. Distance appears to be associated with attendance: Patients in the 21+ km distance group recorded a 57.8% no-show rate, compared with approximately 46–49% among the other distance groups. This suggests that patients travelling substantially farther to the clinic may face additional barriers to attendance.
5. Appointment type shows differences in no-show behaviour: The highest no-show rate was observed for:
   Follow-up - 51.2%
   Diagnostic Test - 49.7%
   Specialist Consultation - 47.4%
   General Consultation - 46.6%

This indicates that appointment type may be relevant when designing targeted attendance interventions.

6. No-show rates vary across age groups: The highest observed rates were among:
   55–64 - 50.8%
   25–34 - 50.7%
   18–24 - 50.2%

The lowest was among the 65+ group at approximately 45.1%.

The differences are relatively moderate, suggesting that age alone may not be sufficient for targeting interventions.

7. Appointment time has relatively small differences. No-show rates were approximately:
   Evening - 49.8%
   Afternoon - 48.4%
   Morning - 48.1%
   The differences are relatively small, suggesting that appointment time may be a weaker driver than previous attendance behaviour or distance.

8. Waiting time did not show a strong distance-based pattern. Average waiting time was approximately 24 minutes across the distance groups. Consequently, Average Waiting Time by Distance to Clinic was not retained as a key dashboard visual, because it did not provide a meaningful distinction between groups.

Business Recommendations: Based on the findings, HealthConnect Clinic should consider the following actions:

- Strengthen reminder coverage: With reminder coverage at 72.7%, a portion of appointments still have no recorded reminder. The clinic should aim to increase reminder coverage, particularly for appointment groups with lower coverage.
- Prioritise patients with previous no-show history: Previous no-show behaviour appears to be a strong indicator of future missed appointments. HealthConnect could introduce additional reminders or confirmation procedures for patients with repeated previous no-shows.
- Provide additional support for patients travelling long distances: Patients in the 21+ km group showed a notably higher no-show rate. The clinic could consider earlier reminders, appointment confirmation, flexible scheduling or other support mechanisms for patients travelling significant distances.
- Review follow-up appointment attendance: Follow-up appointments recorded the highest no-show rate among the appointment types analysed. The clinic should investigate whether follow-up patients require different communication or scheduling approaches.
- Use reminder channels strategically: SMS produced the lowest observed no-show rate among the reminder channels in the dataset. HealthConnect should investigate whether SMS can be incorporated more consistently into its reminder strategy, while recognising that the analysis shows association rather than proof that SMS itself causes lower no-shows.
- Develop targeted rather than blanket interventions: Because some variables showed only modest differences, interventions should focus primarily on stronger behavioural and operational signals rather than treating every patient group identically.

Cross-Track Collaboration: Track: Data Science / Machine Learning

A meaningful dependency between Data Analytics and Data Science/Machine Learning is the use of the Week 5 EDA findings to identify potential predictive features for future modelling.

Information/output exchanged: Data Analytics track identified potentially useful variables and patterns including:

- Previous No-Shows
- Reminder Channel
- Distance Group
- Appointment Type
- Age Group
- Appointment Time
- Appointment Outcome

These findings provide a potential feature set for a future no-show prediction model.

Why the interaction was relevant: The descriptive analysis establishes which variables appear to have useful relationships with appointment attendance. This can help a Data Science/ML track prioritise variables for predictive modelling rather than selecting features without business context.

What changed or improved: The analysis shifted the potential predictive focus toward behavioural history, reminder activity and accessibility factors, particularly previous no-shows and distance, while avoiding unreliable Booking Lead Time data. This creates a clearer foundation for future predictive analysis.

Assumptions, Limitations, Risks & Dependencies:

Issues that remain relevant:

1. Missing values: Waiting Time contains 60 blanks and Distance to Clinic contains 90 blanks. These missing values may reduce the completeness of analyses involving these variables.
2. Demonstration dataset: The dataset appears to contain inconsistencies because it is a demo dataset. Findings should therefore be treated as analytical insights rather than definitive operational conclusions.
3. Reminder data: A lower no-show rate associated with a particular reminder channel does not necessarily mean that the channel directly caused better attendance.

Issues resolved during Week 5:

1. Booking Lead Time: Negative Booking Lead Time values were identified and the variable was excluded from the analysis rather than introducing assumptions or altering questionable source data.
2. Distance analysis: Distance was converted into meaningful groups to support easier comparison of no-show rates.
3. Waiting-time visual: The analysis showed that average waiting time was approximately the same across distance groups. The corresponding visual was therefore removed because it provided limited analytical value.

New issues discovered during Week 5:

1. The main new data-quality issue was the presence of negative Booking Lead Time values.
2. The analysis also highlighted that missing Waiting Time and Distance values need to be considered when interpreting related findings.

Potential impact - These issues may:

1. Reduce the completeness of some analyses.
2. Introduce bias if missing values are not randomly distributed.
3. Affect the reliability of conclusions drawn from the demonstration dataset.
4. Limit the ability to use some variables in future predictive modelling.

How they should be addressed - For future production analysis, HealthConnect should:

1. Validate data-entry rules.
2. Investigate the cause of negative date/time differences.
3. Establish mandatory or controlled fields where appropriate.
4. Monitor missing-value rates.
5. Document how missing values are handled.
6. Validate reminder and appointment outcome records against operational systems.

Week 5 Project Summary (Plans and Completed output): Week 5 was planned to transform the Week 4 foundation into an analytical output by conducting EDA, developing the selected KPIs, identifying meaningful relationships in the HealthConnect appointment data and building an interactive Power BI dashboard.

The following were completed:

- Prepared and reviewed the HealthConnect dataset.
- Conducted focused EDA around appointment attendance and patient support.
- Developed four core KPIs.
- Analysed appointment, patient, reminder, distance, waiting-time and previous no-show patterns.
- Created a three-page Power BI analytical dashboard.
- Developed an Executive Appointment Overview page.
- Developed a No-Show Drivers page.
- Developed a Patient Support & Appointment Operations page.
- Identified key findings and translated them into business recommendations.
- Reviewed and documented data-quality issues and analytical limitations.

Challenges faced - The major challenges encountered included:

- Negative Booking Lead Time values in the demonstration dataset.
- Missing values in Waiting Time and Distance to Clinic.
- Determining which relationships were meaningful enough to include in the dashboard.
- Avoiding unnecessary or repetitive visuals.
- Balancing analytical depth with dashboard readability.

Key decisions: The most important decision was to exclude Booking Lead Time from the Week 5 analysis because of the negative values. Other decisions included prioritising previous no-shows, reminders, distance, appointment characteristics and outcomes while removing visuals that did not provide meaningful variation.

Collaboration: The Week 5 analysis established a potential analytical handoff to the Data Science/Machine Learning track, providing candidate features and business insights that could support future no-show prediction modelling.

Week 6 Focus: Week 6 will focus on building on the Week 5 analytical findings and moving from descriptive analysis toward the next stage of the project, with emphasis on translating insights into a stronger solution, validating findings where possible, and supporting future predictive or AI-driven approaches to reducing missed appointments and improving patient support.

Final assessment of Week 5: Week 5 successfully moved the HealthConnect project from data preparation into business-focused analysis. The resulting dashboard provides management with an interactive view of appointment performance, no-show drivers and patient-support/operational factors. The analysis also establishes a foundation for future predictive modelling and more targeted interventions to reduce missed appointments.